PCB Design Review Checklist

TransEnterix

PCB Part Number and Revision:
IMAGE ENGINE BOARD 8002313 REV 3

Assembled PCB P/N and Revision:
8002277 REV 3

Engineer:
BEN ROY

Date:
7/28/2015

Attendees:
TRX (JEFF, RANDY) + LIC (BEN R, BEN G, VAUGHN)

Part Placement

✓ 1. Polarized components checked
✓ 2. Check the orientation of all connectors
✓ 3. Minimum component body spacing
✓ 4. Bypass capacitors located close to IC power pins
✓ 5. Verify that all series terminators are located near the source
N/A 6. I/O drivers near where their signals leave the board
N/A 7. PCB has ground turrets, power rail test points, and test points for important signals, all labeled
✓ 8. EMI and RFI filtering as close as possible to exit and entry points in shielded areas
✓ 9. Mounting holes electrically isolated or not
✓ 10. Proper mounting hole clearance for hardware
✓ 11. SMD pad shapes checked
N/A 12. Sufficient clearance for socketed ICs
✓ 13. SMD component orientation consistent
✓ 14. Footprints verified against manufacturer datasheet (new parts)

Routing

8/16 4/9

N/A 15. Check for traces running under noisy or sensitive components
✓ 16. Provide multiple vias for high current and/or low impedance traces — ADD SECOND VIA FOR 2.8
✓ 17. Component and trace keepout areas observed
✓ 18. Ground planes where appropriate
✓ 19. Differential lines routed appropriately

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Dimensions

N/A 20. Hole diameter on drawing are finished sizes, after plating. NOTE IN FAB DOCUMENT
✓ 21. Silkscreen legend text weight >=6 mils (USING 4 mil)
✓ 22. Silkscreen legend text height >=30 mils
✓ 23. Pads >=5 mils larger than finished hole sizes ⚠ (NOTE FILLED/PLANATED PADS NOTE IN 3 PLACES)
NA 24. Test pads 200 mils from edge of board
NA 25. Traces >= 15 mils from edge of PCB
✓ 26. Trace width sufficient for current carried
N/A 27. Sufficient clearance for high voltage traces

Silk Screen

✓ 28. No silkscreen text over vias (if vias not soldermasked) or holes
✓ 29. All legend text reads in one or two directions
N/A 30. Company logo / name in silkscreen legend
N/A 31. Date code on PCB
N/A 32. PCB part number on PCB
✓ 33. Assembly part number on PCB
✓ 34. PCB revision on silkscreen legend
N/A 35. Assembly revision blank on silkscreen legend
N/A 36. Serial number blank on silkscreen legend
✓ 37. All silkscreen text located to be readable when the board is populated
✓ 38. All ICs have pin one clearly marked, visible even when chip is installed

Other

✓ 39. DRC and Electrical Test Error Clean
✓ 40. High frequency circuitry precautions observed
N/A 41. Check hole diameters for odd components: rectangular pins, spring pins
✓ 42. Soldermask in relation to via coverage (tenting)
✓ 43. Soldermask swell checked (MAKE NOTE IN FAB DOC)
✓ 44. Fiducials allow for triangulation (multiple sites for flex)
✓ 45. PCB thickness, material, copper weight noted
✓ 46. Thermal reliefs for internal power layers
N/A 47. PCB layout panelized correctly

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Documentation and Release Package

Altium Generated PDF:

✓ A Schematic

Composite drawing:

_____ B One page per metal layer

_____ C One page for Paste, Mask, and Silk (per side) – Paste overlays mask

_____ D Mechanical and Drills

_____ E Used layers only

Items on layout:

✓ A Drill Chart

X B Layer Guide (DEFINED IN FAB DRAWING)

✓ C Title Block

Title Block Includes: N/A

_____ A Title

_____ B Description

_____ C PCB Part Number

_____ D Revision History

_____ E Company

_____ F Address

_____ G Confidentiality Statement

_____ H Engineer

Readme:

✓ A Company Name

✓ B Date

✓ C PCB Part Number

✓ D Description

✓ E Revision

✓ F Contact Information

✓ G Stack Drawing

✓ H Plating/Finish

✓ I RoHS Compliance Statement

✓ J UL Designation

X K Conformance Statement for 6012/6013 and Class ~~(USING~~ IPC-610 CLASS II CALLOUT

✓ L Material Construction (FR4 – etc)
✓ M Any impedance Requirements (IN DRAWING)
✓ N List of all layer filenames and Descriptions
N/A O Stiffener / Flex Circuit Information Explicit

PCB Package Requirements (.zip):

✓ A Correct File name
✓ B Schematic
✓ C Fabrication Drawing
✓ D Gerber Files
N/A E OSB Files
✓ F NC Drill Files
✓ G Assembly drawing
✓ H BOM
✓ I Pick and place files
✓ J README / Fabrication Document
X K 3D Render of PCB
✓ L STEP File
✓ NETLIST

PCB Design Review Checklist

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PCB Part Number and Revision: 8002314
CONSOLE INTERFACE BOARD ~~8002278~~ REV 3

Assembled PCB P/N and Revision:
8002278

Engineer:
BEN ROY

Date:
7/28/2015

Attendees:
TRX (JEFF, RANDY) + LIC (BENR, BENG, VAUGHN)

Part Placement

✓ 1. Polarized components checked
✓ 2. Check the orientation of all connectors
✓ 3. Minimum component body spacing
✓ 4. Bypass capacitors located close to IC power pins
✓ 5. Verify that all series terminators are located near the source
N/A 6. I/O drivers near where their signals leave the board
✓ 7. PCB has ground turrets, power rail test points, and test points for important signals, all labeled
✓ 8. EMI and RFI filtering as close as possible to exit and entry points in shielded areas
✓ 9. Mounting holes electrically isolated or not
✓ 10. Proper mounting hole clearance for hardware
✓ 11. SMD pad shapes checked
N/A 12. Sufficient clearance for socketed ICs
✓ 13. SMD component orientation consistent
✓ 14. Footprints verified against manufacturer datasheet (new parts)

(1) ✓ ADD TERMINATING RESISTORS TO CLOCKS & DATA 0201's

(2) ✓ CHECK TP FOOTPRINTS

Routing

(3) ✓ ADD SMD 2.8VIA 1.5

✓ 15. Check for traces running under noisy or sensitive components
✓ 16. Provide multiple vias for high current and/or low impedance traces
✓ 17. Component and trace keepout areas observed
✓ 18. Ground planes where appropriate
✓ 19. Differential lines routed appropriately

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Dimensions

✓ 20. Hole diameter on drawing are finished sizes, after plating.
✓ 21. Silkscreen legend text weight >=6 mils (4mil)
✓ 22. Silkscreen legend text height >=30 mils
✓ 23. Pads >=5 mils larger than finished hole sizes
✓ 24. Test pads 200 mils from edge of board
✓ 25. Traces >= 15 mils from edge of PCB
✓ 26. Trace width sufficient for current carried
N/A 27. Sufficient clearance for high voltage traces

Silk Screen

✓ 28. No silkscreen text over vias (if vias not soldermasked) or holes
✓ 29. All legend text reads in one or two directions CHECK ORIENTATION (4) ✓
N/A 30. Company logo / name in silkscreen legend
N/A 31. Date code on PCB (5) MOVE PT # TO SILK
N/A 32. PCB part number on PCB
✓ 33. Assembly part number on PCB
N/A 34. PCB revision on silkscreen legend
N/A 35. Assembly revision blank on silkscreen legend
N/A 36. Serial number blank on silkscreen legend
✓ 37. All silkscreen text located to be readable when the board is populated
✓ 38. All ICs have pin one clearly marked, visible even when chip is installed
✓ (6) ADD PIN 1 TO CONN J3

Other

✓ 39. DRC and Electrical Test Error Clean
✓ 40. High frequency circuitry precautions observed
N/A 41. Check hole diameters for odd components: rectangular pins, spring pins
N/A 42. Soldermask in relation to via coverage (tenting) (7) NOTE IN FAB DOC
43. Soldermask swell checked
✓ 44. Fiducials allow for triangulation (multiple sites for flex)
✓ 45. PCB thickness, material, copper weight noted
N/A 46. Thermal reliefs for internal power layers
N/A 47. PCB layout panelized correctly

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Documentation and Release Package

Altium Generated PDF:

- ✓ A Schematic
- *Composite drawing:*
- ✓ B One page per metal layer
- ✓ C One page for Paste, Mask, and Silk (per side) – Paste overlays mask
- ~~D Mechanical and Drills~~
- ~~E Used layers only~~

Items on layout:

- ✓ A Drill Chart
- X B Layer Guide (IN FAB DOC)
- ✓ C Title Block

Title Block Includes: N/A

- ~~A Title~~
- ~~B Description~~
- ~~C PCB Part Number~~
- ~~D Revision History~~
- ~~E Company~~
- ~~F Address~~
- ~~G Confidentiality Statement~~
- ~~H Engineer~~

Readme:

- ✓ A Company Name
- ✓ B Date
- ✓ C PCB Part Number
- ✓ D Description
- ✓ E Revision
- ✓ F Contact Information
- ✓ G Stack Drawing
- ✓ H Plating/Finish
- ✓ I RoHS Compliance Statement
- ✓ J UL Designation
- ✓ K Conformance Statement for 6012/6013 and Class

IPC-A-610 II

_____L Material Construction (FR4 – etc)
_____M Any impedance Requirements
_____N List of all layer filenames and Descriptions
_____O Stiffener / Flex Circuit Information Explicit

PCB Package Requirements (.zip):

_____A Correct File name
_____B Schematic
_____C Fabrication Drawing
_____D Gerber Files
_____E OSB Files
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